

WORK ORDER PLATFORM – PRODUCT REQUIREMENTS

0. Document control

Field	Value
Document ID	PRD-WRK-001
Feature name	Work Order Platform v1 – intake to collection
Module	WRK (with QUO, PAY, OBL, REC sub-areas)
Status	Draft
Version	0.1
Author	J. Brown / Claude (drafted)
Product owner	Jordan Brown
Business owner	Jordan Brown
Engineering lead	TBD
Target release	R1 Pilot (~5 users, "Incoming WOs" live)
Last updated	2026-08-07

Reviewers and sign-off

Role	Name	Reviewed	Approved	Date
Business owner	Jordan Brown	☐	☐	
Operations lead		☐	☐	
Engineering lead		☐	☐	
Finance (touches money)		☐	☐	

1. Summary

Seamless FM runs ~21,000 work orders through a ClickUp workspace plus five disconnected side tools, with every work order re-keyed on intake, assigned twice (ClickUp and a Teams PDF channel), quoted in a separate tool, paid through another, and audited in a third. The One is a single operations platform that carries a work order natively from intake through dispatch, quoting, approval, fulfilment, technician payment, completion audit, invoicing, and collection – with an internal-versus-client visibility boundary on every update and a clock on everything the company owes. It replaces ClickUp and absorbs the side tools, so nothing about a work order lives outside the system of record.

2. Problem and opportunity

2.1 Problem statement

Operations staff run every work order across ClickUp plus Yoda (quotes), PPR (tech payments), Argus (completion audit), Quotato (tech texts/calls), a Teams PDF channel (assignment), and manual re-entry into each client's CMMS. Nothing connects: a dispatcher cannot see from one place what stage a job is in, what has been promised to the client, what the tech was told, whether the tech was paid, or whether the invoice went out. Deadlines live in people's heads – a quote that should have gone out Tuesday surfaces when the client calls angry. Finished jobs sit un-invoiced because nobody notices they cleared audit.

2.2 Evidence

Evidence	Source	Value
Work orders in the live workspace	ClickUp workspace measurement, 2026-07-23	~21,239
Workspace users	Same measurement	98
Lists a WO can live in (routing surface)	Same measurement	163
WOs stuck awaiting client approval, single sample	Same measurement	46
Distinct custom fields staff maintain by hand	Same measurement	102
Side tools that each hold part of the WO record	Repo inventory	5 (Yoda, PPR, Argus, Quotato, escalation bot)
Duplicate assignment channels	<code>product/wo-lifecycle.md</code> stage 3	2 (ClickUp + Teams PDF)

2.3 Cost of doing nothing

Every WO continues to cost duplicate keying at intake, assignment, TL Rep, and invoicing. Missed obligations keep surfacing as client escalations rather than internal warnings. The completion-audit backlog keeps delaying invoicing, and collection ages unmeasured. The fallback is licensing Facilio at roughly \$25K/yr – while still hand-building every integration, per its known weaknesses.

2.4 Business objective and success metrics

#	Metric	Definition	Baseline	Target	Measured by	By when
M1	Intake double-entry	Number of systems a new WO is manually keyed into (platform, ClickUp, Teams)	3	1	Pilot observation	R1 exit
M2	Obligation breach rate	Breached obligations ÷ obligations opened, per week (BR-OBL-007...013)	Not measurable today – instrumented from pilot day 1	Baseline – 50% within 8 weeks of pilot	Platform report	R2 entry
M3	Approval-chase coverage	Quotes awaiting client approval > 5 business days that have a recorded follow-up (BR-OBL-011)	Unknown; 46 WOs stuck in one sample with no tracked follow-up	100%	Platform report	R1 exit
M4	Quote review turnaround	Median business hours from quote submission to ATL decision (BR-OBL-009)	Not measured today	≤ 4 business hours	Platform report	R1 exit
M5	Audit-to-invoice lag	Median days from final soft close to invoice sent	Not measurable in ClickUp – instrumented from pilot	Baseline – 30%	Platform report	R2 exit

2.5 Counter-metrics

- **Quote quality must not drop.** Rejection rate at ATL review must not rise above its first-month baseline while turnaround tightens.
- **Technician payment speed must not slow.** Median request-to-paid time must stay ≤ 2 business days (BR-OBL-012); techs go where they get paid.
- **Client-visibility incidents = 0.** No internal content may reach a client surface (BR-ACC-001); one incident is a stop-the-line event.

3. Users and personas

Persona	Role in this feature	Job to be done	Current pain	Frequency of use	Device / context
Dispatcher (OM / Senior OM)	Primary	Source a tech, run both visits, quote, soft-close my book	5 tools + Teams pings; deadlines in my head	All day	Desktop, office
Team Lead / Assistant TL	Primary	Accept work, audit quotes, keep the book honest	Quote review arrives by ping; no queue, no clock	Hourly	Desktop
Operations Admin (OA)	Primary	Intake WOs; push curated updates to client CMMS (TL Rep)	Re-keys everything twice	All day	Desktop
Account Manager	Primary	Accept/decline; own the client; chase approvals	Can't see what the client was told without asking	Daily	Desktop, calls
AP team	Primary	Pay techs fast and correctly; order parts	Requests arrive as links in chat (PPR); no queue	Daily	Desktop
AR team (HAR / Senior / AR)	Primary	Audit finished WOs, invoice, collect	Audit checklist lives in a separate tool; aging invisible	Daily	Desktop
Executive guest	Secondary	See rollups without touching records	Reads exported snapshots	Weekly	Desktop, read-only
Technician	External	Get the job, send photos, get paid	Texts vanish into one dispatcher's phone	Per job	SMS/phone only – no login in v1
Client / FM contact	External	Approve quotes, see curated status	Updates arrive inconsistently per CMMS	Per WO	Their own CMMS – no login in v1
Automations (n8n, bots)	Service	Act on the platform under attribution	No first-class identity in ClickUp	Continuous	API (BR-IAM-003)

4. Scope

4.1 In scope

- Work-order record and lifecycle: single status pipeline, phases, books/routing, accept-decline, activity history (BR-WRK-001...009)
- Internal-vs-client visibility classification on every update (BR-ACC-001...003)
- Technician message/call thread on the WO (Quo channel, read view)
- Quote building, review, approval, sending – with incurred/proposed structure and money math (BR-QUO-001...010)
- Technician payment requests and the AP decision flow (BR-PAY-001...005)
- Obligations: the seven clocks, tiers, snooze, evidence-based resolution, notifications (BR-OBL-001...013)
- Completion audit and invoicing pipeline: grey-flag checks, release gate, ready → invoiced → paid, collections view (BR-REC-001...012)

- Work-order list, detail, Pulse (triage), Receivables screens; KPI figures (§13)
- Roles, tiers, and permission gates per §9

4.2 Explicitly out of scope

Not included	Why	Revisit
Vendor database, pooled map, my-book lens	Own module; dispatch works from existing contacts in v1	R1, PRD-VND-001
Live client-CMMS adapters (Ecotrak first) and email-to-WO ingestion	Feasibility spike pending; TL Rep stays manual	R1-R2, PRD-INT-001
Automations rules engine, saved views, dashboards beyond KPI row	Pilot doesn't need them; n8n covers automation externally	R2
Client portal, vendor portal, technician login	Different trust levels and support paths — deliberate later phase	R3
QuickBooks sync, client invoicing documents	Invoicing recorded in-platform v1; documents produced as today	R2, PRD-FIN-001
Preventive maintenance, assets	<code>pm_scheduled</code> status exists; no PM engine	R3
One-time ClickUp import (21K WOs)	Cutover concern, not v1	R2

4.3 Assumptions

#	Assumption	Confirmed by	Status
A1	The 19-status pipeline (Catalog Appendix A) is complete and correct for all teams	Jordan Brown	Confirmed in prototype seed
A2	Quote gates: build = Senior OM+, approve & send = ATL+	Jordan Brown	Confirmed 2026-07-30 (<code>product/quotes-payments.md §4</code>)
A3	OT = ×1.5; tax derives from site state; options are internal	Jordan Brown	Confirmed 2026-07-30
A4	Business hours are Mon-Fri 08:00-18:00 CT for all obligation clocks	Jordan Brown	Confirmed in escalation rulebook
A5	~5 named pilot users will run "Incoming WOs" on the platform for R1	Jordan Brown	Pending
A6	Technicians remain phone/SMS-only in v1 — no tech-facing screens	Jordan Brown	Confirmed

4.4 Dependencies

#	Depends on	Type	Owner	Needed by	Status
D1	Business Rules Catalog v1.0 approved	Internal	Jordan Brown	Build start	Drafted, pending review
D2	Quo/OpenPhone data feed (calls, transcripts, SMS) for the Messages view	Internal (Quotato)	Jordan Brown	R1	Prototype uses mirrored data
D3	Ecotrak staging access for the first intake adapter	Third-party	Jordan Brown	R1(S8)	Credentials in hand

#	Depends on	Type	Owner	Needed by	Status
D4	Corrigo / ServiceChannel API feasibility spike	Third-party	TBD	R2 scoping	Not started
D5	Real payment-method list from AP	Internal	AP lead	Build	Open — free text until then

5. Business process — current and target

5.1 Current process (as-is)

A WO appears on a client CMMS, in email, or in Seamgo. An OA re-keys it into ClickUp, and in parallel posts it as a PDF to a Teams channel where TLs tag dispatchers — two assignment channels for one job. The dispatcher finds a tech from a pooled map plus their own book, texts and calls them (the thread lives in one person's phone unless Quotato catches it), and sends them to assess. The quote is built in Yoda, audited by a TL/ATL informally, and re-keyed into the client's CMMS by the OA ("TL Rep"). Waiting on approval, nobody owns the follow-up — WOs sit for weeks. After fulfilment, the dispatcher soft-closes: photos, checkmarks, and a PPR link for the tech's payment. AR audits in Argus's shadow, invoices into the client CMMS, and chases collection from memory. **Workarounds that are really requirements:** the Teams channel exists because ClickUp assignment isn't noticed (→ obligations/notifications); the escalation bot exists because deadlines had no owner (→ the OBL rules); Argus exists because invoicing errors were caught too late (→ the REC gate).

5.2 Target process (to-be)

Step	Actor	Action	System behaviour	Business rules	Handoff to
1	OA	Registers an incoming WO (source, client, site, trade, NTE, SLA date, description)	Creates the record; starts intake clocks; emergency status starts the 2-hour clock	BR-INT-003, BR-OBL-007	TL/ATL/AM
2	TL / ATL / AM	Accepts or declines with reason	Accept admits it to the pipeline; decline records decider + reason and ends the flow	BR-WRK-004	OA
3	OA / TL	Routes the WO to a dispatcher's book	Home book changes; activity history records it; dispatcher notified	BR-WRK-003	Dispatcher
4	Dispatcher	Sources a tech; contacts via SMS/call	Thread captured on the WO's Messages view; never client-visible	BR-DSP-001, BR-ACC-002	Technician
5	Dispatcher	Schedules and runs the assessment visit	Statuses <code>assessment scheduled</code> → <code>assessment ongoing</code> ; quote clock starts at <code>waiting for quote</code>	BR-WRK-005, BR-OBL-008	Technician
6	Dispatcher (Senior OM)	Soft close pt 1: before-photos + builds the quote (incurred + options)	Totals, tax, NTE check computed live; submit moves quote to pending approval	BR-WRK-006, BR-QUO-001...006, BR-QUO-010	ATL
7	ATL / TL	Reviews the quote; approves & sends, or rejects with feedback	Review clock (4 bus. hrs); reject returns to draft with feedback on the WO	BR-OBL-009, BR-QUO-007...009, BR-ACC-005	OA
8	OA	TL Rep: pushes the client summary + status to the client's CMMS	Only client-visible content leaves; push recorded	BR-ACC-001, BR-ACC-006, BR-INT-002	Client
9	AM / OA	Waits on client, follow-up every 5 business days; or records on-site approval	Approval-followup clock; on-site approval skips the wait	BR-OBL-011, BR-WRK-009	Dispatcher

Step	Actor	Action	System behaviour	Business rules	Handoff to
10	AP	Orders parts when needed	please order parts → waiting for parts ; parts noted on the WO	–	Dispatcher
11	Dispatcher	Schedules and runs the fulfilment visit (same tech first)	Schedule clock (2 bus. hrs from approval); job scheduled → job ongoing	BR-OBL-010, BR-DSP-003	Technician
12	Dispatcher	Soft close final: after-photos, checklist, raises tech payment request	Completeness checks; request enters AP queue with its 2-business-day clock	BR-WRK-007, BR-PAY-001, BR-PAY-004	AP
13	AP	Decides the payment request; pays	requested → approved → paid, attributed; cost lands on the WO	BR-PAY-002...003, BR-FIN-002	AR
14	AR	Completion audit: automatic checks + reviewer confirmation	Findings computed (photos, sign-off, quote, CICO, CO#); release gate	BR-REC-001...010	AR
15	AR	Invoices; collections works aging until paid	ready → invoiced → paid; every follow-up recorded	BR-REC-011...012	–

5.3 What changes for whom

Persona	Stops doing	Starts doing	Change impact
OA	Keying into ClickUp + posting Teams PDFs	Keying once into the platform	Medium
Dispatcher	Yoda, PPR links, memory-driven deadlines	Quote and payment on the WO; working from Pulse	High
TL/ATL	Ad-hoc quote review on ping	Working a review queue with a 4-hour clock	Medium
AP	Processing PPR links from chat	Working a payment queue with a 2-day clock	Medium
AR	Argus side-tool + manual aging	In-platform audit queue and invoicing lanes	Medium
AM	Chasing approvals from memory	Prompted follow-ups every 5 business days	Low

6. Functional requirements

Priority: **Must** / **Should** / **Could** / **Won't-this-release**.

6.1 Work-order record & lifecycle – WRK

ID	Requirement	Persona	Priority	Business rules	Acceptance criteria
FR-WRK-001	A user can register a work order with source, client, FM company, billing entity, site (city/state), trade, priority, NTE, SLA due date, date received, and description	OA	M	BR-INT-003	Given a new WO from any of the five sources, when the OA saves it with mandatory fields complete, then it exists with a unique WO number and appears in the intake view
FR-WRK-002	A TL, ATL, or AM can accept or decline a registered WO, with a reason required on decline	TL/ATL/AM	M	BR-WRK-004	Given a registered WO, when an AM declines it without a reason, then the decline is refused; with a reason, the WO leaves the pipeline and records decider, reason, time

ID	Requirement	Persona	Priority	Business rules	Acceptance criteria
FR-WRK-003	A user can change a WO's status along the single pipeline, and the phase bar reflects the status's phase	All internal	M	BR-WRK-001, BR-WRK-002	Given a WO in <code>waiting for quote</code> , when its status changes to <code>quote ready</code> , then the activity history records actor and both statuses, and the phase shows Quote
FR-WRK-004	A user can route a WO to a dispatcher's book; the WO has exactly one home book	OA/TL	M	BR-WRK-003	Given a WO homed in Book A, when it is routed to Book B, then Book B is its only home and the change is attributed in the history
FR-WRK-005	The WO detail shows people (dispatcher, AM), site, dates, money, flags, photos, checklist, obligations, and full field set	All internal	M	BR-FIN-001	Given the hero demo WO, when opened, then all listed panels render with its real values
FR-WRK-006	A user can post an update on a WO classified internal or client-visible at creation, immutable afterwards	All internal	M	BR-ACC-001	Given a WO, when a user posts an update marked client-visible, then it is flagged as such in the feed and only such updates are offered to any outbound push
FR-WRK-007	A user can upload photos to a WO, classified before / after / other	Dispatcher	M	BR-WRK-006, BR-WRK-007	Given an assessment done, when the dispatcher uploads two photos tagged "before", then soft-close pt 1's photo condition is satisfied
FR-WRK-008	The WO carries a soft-close checklist for part 1 and final, and shows completeness	Dispatcher	M	BR-WRK-006, BR-WRK-007	Given final soft close with after-photos missing on a non-BFI WO, when the dispatcher views the checklist, then the missing item is shown as blocking
FR-WRK-009	Every write to a WO appears in an append-only, filterable audit-trail view	TL/ATL, AR	M	BR-IAM-002	Given any change, when the audit trail is opened, then the change appears with actor, timestamp, and before/after
FR-WRK-010	A user can cancel/postpone a WO with a recorded reason	TL/ATL/AM	M	BR-WRK-008	Given an active WO, when cancelled without a reason, then the change is refused; with one, status becomes <code>!! canceled/postponed</code> off-pipeline
FR-WRK-011	A user can record an on-site approval (who granted it, when) which skips the client-approval wait	Dispatcher	S	BR-WRK-009	Given a WO assessed with on-site approval, when recorded, then the WO may move directly to scheduling and no approval-followup clock starts
FR-WRK-012	The technician SMS/call thread (calls, transcripts, summaries, messages, media) is visible on the WO	Dispatcher, TL	M	BR-ACC-002	Given the demo WO with 2 calls and 5 messages, when Messages opens, then the thread renders chronologically and offers no client-visible marking

6.2 Work-order list & search

ID	Requirement	Persona	Priority	Business rules	Acceptance criteria
FR-WRK-020	The WO list shows number, title, client, site, trade, entity, NTE, priority, status, age, and obligation state per row, filterable by status group and status, searchable	All internal	M	BR-WRK-002	Given the seeded set, when filtered to group "done", then only done-group WOs appear with correct counts

ID	Requirement	Persona	Priority	Business rules	Acceptance criteria
FR-WRK-021	The list can sort by obligation urgency ("sort by breach")	Dispatcher	S	BR-OBL-003	Given WOs with tier 3, 1, 0 obligations, when sorted by breach, then they order 3 → 1 → 0

6.3 Quotes – QUO

ID	Requirement	Persona	Priority	Business rules	Acceptance criteria
FR-QUO-001	A Senior OM+ can create/edit the WO's quote: an incurred section plus one or more proposed options, each with narrative and typed line items (service, labor, part, material)	Senior OM	M	BR-ACC-004, BR-QUO-001, BR-QUO-002	Given a senior_om actor, when they add Option A with 3 lines, then totals compute live; given an om actor, then editing is refused and the controls show locked with the reason
FR-QUO-002	Line amounts, option totals, tax, and grand total are computed, never typed	Senior OM	M	BR-QUO-003...005, BR-FIN-005	Given a labor line 4 × \$120 with OT, when saved, then its amount reads \$720; no amount field is directly editable
FR-QUO-003	The quote shows the money position alongside: cost, grand total, tax, NTE, profit, margin, with an NTE meter	Senior OM, ATL	M	BR-FIN-001...004, BR-QUO-010	Given a grand total above NTE, when displayed, then the NTE state is visibly flagged to builder and approver
FR-QUO-004	A client-facing summary is generated from the included proposed sections and is editable before sending	Senior OM, ATL	M	BR-QUO-006	Given two options with one marked for summary, when the summary generates, then only that option's content appears, and edits persist
FR-QUO-005	Quote flows draft → pending approval → approved → sent; submit requires Senior OM+, decisions require ATL+	Senior OM, ATL	M	BR-QUO-007...008, BR-ACC-005	Given a pending quote and an atl actor, when approved and sent, then state is sent and both decisions are attributed; an om attempting any transition is refused
FR-QUO-006	Rejection returns the quote to draft with the reviewer's feedback recorded on the WO	ATL	M	BR-QUO-009	Given a pending quote, when rejected with feedback, then the builder sees the feedback on the WO and the quote is editable again as a new revision
FR-QUO-007	A quotes index lists every quote with WO, state, total, and last-touched	TL/ATL	S	—	Given three quotes in mixed states, when the index opens, then all three appear with correct states and totals

6.4 Technician payments – PAY

ID	Requirement	Persona	Priority	Business rules	Acceptance criteria
FR-PAY-001	A user can raise a payment request on a WO: payee (registered vendor or manual name + phone), amount, purpose, method, optional alternate recipient with reason	Dispatcher, AP	M	BR-PAY-001, BR-PAY-005	Given a WO, when a request is raised with a manual payee and no phone, then it is refused; complete, it enters the AP queue as requested
FR-PAY-002	Prior payments and requests on the WO are shown when raising a new one	Dispatcher	M	—	Given a WO with one paid request, when a second is raised, then the first is visible with amount and state before submission
FR-PAY-003	AP can approve, reject, or mark paid; every decision attributed; paid amounts land in WO cost	AP	M	BR-PAY-002...003, BR-FIN-002	Given a requested payment, when AP marks it paid, then WO cost includes it and profit/margin update

ID	Requirement	Persona	Priority	Business rules	Acceptance criteria
FR-PAY-004	Amount entry is hardened: currency strings parse, nonsense is refused	Dispatcher	S	—	Given "1,250.00" entered, then the amount records as 1250.00; given "12oo", then it is refused with a message

6.5 Obligations & notifications – OBL

ID	Requirement	Persona	Priority	Business rules	Acceptance criteria
FR-OBL-001	The system derives obligations from WO/quote/payment state per the seven clock rules, with business-hours or 24×7 clocks as each rule specifies	System	M	BR-OBL-001...002, BR-OBL-007...013	Given a WO entering waiting for quote Friday 17:00, when 2 business days elapse, then the obligation breaches Tuesday 17:00 (clock paused over the weekend)
FR-OBL-002	Obligations expose tier 0–3 per the thresholds; tier changes notify once per tier	All internal	M	BR-OBL-003, BR-OBL-006	Given an obligation crossing 80%, then exactly one due-soon notification is ever produced for it, even across restarts
FR-OBL-003	Pulse groups a user's obligations: Needs me now (tier 2–3), Due soon (tier 1), Watching (tier 0)	All internal	M	BR-OBL-003	Given seeded obligations at known tiers, when Pulse opens, then each sits in the correct column with its clock
FR-OBL-004	A user can snooze an obligation up to 72h with a mandatory reason; there is no dismiss anywhere	All internal	M	BR-OBL-004...005	Given an obligation, when snoozed without a reason or beyond 72h, then it is refused; no interface offers dismissal
FR-OBL-005	Obligations resolve automatically on evidence of the owed action	System	M	BR-OBL-004	Given a breached quote-owed obligation, when the quote is submitted, then the obligation resolves without user action
FR-OBL-006	A notification inbox lists pings with read/read-all; WO rows and the WO detail surface their open obligations	All internal	M	BR-OBL-006	Given three unread pings, when read-all is used, then the unread count is zero and pings remain listed

6.6 Completion audit & receivables – REC

ID	Requirement	Persona	Priority	Business rules	Acceptance criteria
FR-REC-001	Finished WOs enter an audit queue showing, per WO, every automatic check's outcome (pass / minor / major / n-a / offline) – not only failures	AR	M	BR-REC-001, BR-REC-003...009	Given a WO missing its sign-off sheet, when its row expands, then the SO check reads major and the passing checks read pass
FR-REC-002	The reviewer records the four audit marks (GTG, Elise, Admin, Quote); release requires Admin + Quote and no blocking findings	AR	M	BR-REC-002, BR-REC-010	Given a WO with zero major findings and both release checks ticked, then it moves to the invoicing Ready lane; with a major finding, release is refused
FR-REC-003	Findings carry resolution paths pointing at the WO or the external store holding the evidence	AR	S	—	Given a CO#-format finding, when expanded, then it links to where the CO# is corrected
FR-REC-004	The invoicing view shows Ready → Invoiced → Paid lanes with amount, invoice number, and aging	AR	M	BR-REC-011...012	Given a released WO, when invoiced is recorded, then it appears in Invoiced with aging counting from that date

ID	Requirement	Persona	Priority	Business rules	Acceptance criteria
FR-REC-005	An unresolved-audit WO cannot be marked invoiced	AR	M	BR-REC-001	Given a WO still in the audit queue, when an invoice is attempted, then it is refused with the blocking findings listed

6.7 Reporting & KPIs – RPT

ID	Requirement	Persona	Priority	Business rules	Acceptance criteria
FR-RPT-001	The KPI row shows: active WOs; awaiting client approval (count + oldest age); ready to invoice (count + amount); margin % and average profit	TL, Exec	M	BR-FIN-003...004	Given the seeded set, when the list loads, then each figure matches a hand count of the same data
FR-RPT-002	Executive guests can view KPI figures and lists read-only	Exec	S	BR-ACC-007	Given an exec actor, when any edit is attempted, then it is refused

7. Business rules applied

Full text in `BUSINESS-RULES-CATALOG-v1.md`. This PRD applies: **IAM** 001-004 · **ACC** 001-007 · **WRK** 001-009 · **QUO** 001-010 · **FIN** 001-006 · **PAY** 001-005 · **OBL** 001-013 · **DSP** 001-003 · **REC** 001-012 · **INT** 001-003. All are **new in catalog v1.0**, drawn from existing practice; none conflict with a prior version.

8. Data requirements

8.1 Entities touched

Entity	Created	Read	Updated	Deleted	Owning module
Work order	✓	✓	✓	Soft only (cancel)	WRK
Update / comment (with visibility flag)	✓	✓	— (immutable)	X	WRK
Photo / attachment	✓	✓	X	X	WRK
Status (shared pipeline)	X	✓	X	X	WRK
Book (routing container)	X	✓	✓ (membership)	X	WRK
Technician thread (calls, messages)	via feed	✓	X	X	WRK
Quote (sections, lines, revisions)	✓	✓	✓	X (revisions retained)	QUO
Payment request	✓	✓	✓ (state only)	X	PAY
Vendor (minimal payee record)	✓	✓	✓	X	PAY
Obligation / notification	System	✓	System + snooze	X	OBL
Audit marks & findings	✓	✓	✓	X	REC
Activity entry	System	✓	X (append-only)	X	IAM
Principal (person / service account)	✓	✓	✓	Deactivate only	IAM

8.2 Key fields (work order)

Field	Type	Required	Default	Validation	PII	Retention
WO number	Identifier	Yes	System-issued	Unique	No	Life of company (Q7)
External ref (client CMMS #)	Text	No	—	—	No	Same as WO
Client / FM company / billing entity	Reference	Yes	—	Known values	No	Same as WO
Site: city, state	Text	Yes	—	State drives tax (BR-QU0-005)	No	Same as WO
Trade	Reference	Yes	—	Known trades	No	Same as WO
Priority	Enum	No	normal	urgent/high/normal/low	No	Same as WO
NTE	Money	No	—	≥ 0	No	Same as WO
SLA due date	Date	No	—	—	No	Same as WO
Status	Reference	Yes	Open	Pipeline only (BR-WRK-001)	No	Same as WO
Description	Rich text	Yes	—	—	Possible (site contacts)	Same as WO
Payee name + phone (manual payments)	Text	When manual	—	Phone format	Yes	Q7

8.3 Data ownership and lifecycle

- **Source of truth:** The One, from acceptance onward. The client's CMMS remains the source for the client's own WO identity (external ref). ClickUp is historical only (BR-INT-001).
- **Retention:** work orders and money records — indefinitely pending a policy decision (Q7). Technician thread media — same as the WO it attaches to.
- **Deletion behaviour:** nothing hard-deletes. Cancellation is a status (BR-WRK-008); principals deactivate; history is immutable (BR-IAM-002).
- **Tenant scoping:** single company, multi-brand. Every WO carries its billing entity; client-visibility (BR-ACC-001) is the isolation boundary that matters.

9. Permissions and visibility

[illegible]

Persona	Create WO	View	Edit WO	Accept/Decline	Quote build	Quote approve/send	Payment raise	Payment decide	Audit release	Record scope
AM	✓	✓	✓	✓	✓	✓	✓	X	X	Their clients (view all)
AP	X	✓	X	X	X	X	✓	✓	X	All
AR (by tier)	X	✓	X	X	X	X	X	X	✓ (waivers Senior+, BR-REC-010)	All
Exec guest	X	✓ (rollups)	X	X	X	X	X	X	X	Read-only
Admin	✓	✓	✓	✓	✓	✓	✓	✓	✓	All
Service accounts	Per grant	Per grant	Per grant	X	X	X	Per grant	X	X	Per grant (BR-IAM-003)

Presentation rule: a permitted-but-gated action renders locked with the reason – never hidden (`product/quotes-payments.md` §3).

Field-level restrictions

Field	Hidden from	Reason
Cost, technician payments, profit, margin	Technicians, clients (all outbound surfaces)	BR-ACC-003 – commercially sensitive
Internal updates, technician thread	All client-facing channels	BR-ACC-001...002
Tier-specific gates within OM/AR ladders	—	Defined per tier at R1 RBAC (Q3)

10. Workflow and approvals

10.1 States

Work order: the 19-status pipeline (Catalog Appendix A). Quote: draft / pending approval / approved / sent (BR-QUO-007). Payment request: requested / approved / paid / rejected (BR-PAY-003). Obligation: open / snoozed / resolved (BR-OBL-004...005). Invoicing: ready / invoiced / paid (BR-REC-011).

10.2 Transitions (governing rules)

From	To	Trigger	Who	Conditions	Rules
Registered	In pipeline	Accept	TL/ATL/AM	—	BR-WRK-004
Registered	Declined (out)	Decline	TL/ATL/AM	Reason recorded	BR-WRK-004
Quote draft	Pending approval	Submit	Senior OM+	Quote has ≥ 1 proposed section	BR-ACC-004, BR-QUO-007
Pending approval	Approved → Sent	Approve & send	ATL+	NTE flag surfaced if exceeded	BR-ACC-005, BR-QUO-008, BR-QUO-010

From	To	Trigger	Who	Conditions	Rules
Pending approval	Draft	Reject	ATL+	Feedback recorded	BR-QUO-009
!! waiting for approval	approved	Client approves (or on-site recorded)	OA / dispatcher	On-site approval attributed	BR-WRK-009
Payment requested	Approved → Paid	AP decision	AP only	Attributed	BR-PAY-002...003
Audit queue	Invoicing Ready	Release	AR	Admin + Quote ticked, no majors	BR-REC-002, BR-REC-010
Ready	Invoiced → Paid	Invoice sent / payment received	AR	Aging from invoice date	BR-REC-011...012

10.3 Approval routing

Condition	Approver	Sequence	SLA	On breach	Delegation allowed
Quote submitted	ATL (TL/AM/admin may act)	Single	4 business hours	Tiered obligation escalation (BR-OBL-009, BR-OBL-003)	Any ATL+ may act – pooled, not personal
Payment requested	AP	Single	2 business days	Tiered obligation escalation (BR-OBL-012)	Within AP
Audit release	AR reviewer (waivers Senior AR+)	Single	– (grey-flag aging monitored)	Aging visible in queue	Within AR per tier

- **Approver on leave:** approvals are pooled by role, not personal queues; anyone holding the role may act.
- **Self-approval:** unresolved – Q4 (an ATL who builds a quote approving their own work; conflict register C1).
- **Revocability:** a sent quote is not revocable; changes create a new revision (BR-QUO-001). A paid payment is not reversible in-system v1(Q8).
- **Audit trail:** every decision records actor, timestamp, and outcome (BR-IAM-002).

11. Integrations

System	Direction	Trigger	Data exchanged	Frequency	On failure	Owner
Client CMMS – Ecotrak first; Corrigo, ServiceChannel later	Inbound (v1: manual entry; R1: adapter)	New WO / status change at client	WO identity, scope, NTE, SLA	Per event	OA keys manually – the platform never blocks intake on an adapter	Jordan Brown
Client CMMS (TL Rep)	Outbound (v1: manual; R2: push)	Quote sent / curated update	Client-visible summary + status only (BR-INT-002)	Per event	OA pushes manually; the WO records what was sent either way	Jordan Brown
Quo / OpenPhone (via Quotato)	Inbound	Tech call or SMS	Calls, transcripts, summaries, messages, media onto the WO	Near-real-time	Thread gap flagged; dispatcher phones the tech – dispatch never blocks	Jordan Brown

System	Direction	Trigger	Data exchanged	Frequency	On failure	Owner
SharePoint (deliverables store)	Read	Completion audit runs	Folder contents: sign-off, before/after photos, labels	Per audit	Checks report "offline – skipped" honestly; audit proceeds on the remaining checks; release still requires the reviewer	Head of AR
n8n automations	Bidirectional	Platform events	Per automation, as a service account	Continuous	Automation queues retry; obligations are the human backstop	Jordan Brown
ClickUp (legacy)	None – read-only, forever	–	Historical reference only	–	– (BR-INT-001)	Jordan Brown

12. Notifications and communications

Event	Recipient	Channel	Timing	Content summary	Can user opt out
Obligation reaches due-soon (tier 1)	The owing role/user	In-app (bell + Pulse)	At 80% of allowance, once (BR-OBL-006)	What's owed, on which WO, time left	No – snooze with reason instead
Obligation breaches (tier 2)	The owing role/user	In-app	At 100%, once	What breached, by how much	No
Obligation critical (tier 3)	Owner + their lead	In-app	At 200% or breached emergency, once	Same + escalation	No
Emergency WO arrives	Dispatch + TL	In-app	Immediate, clock visible (BR-OBL-007)	WO, site, 2-hour clock	No
Quote decided (approved/rejected)	The builder	In-app	Immediate	Outcome + feedback if rejected	No
Payment decided	The requester	In-app	Immediate	Outcome + method	No
WO routed to my book	Dispatcher	In-app	Immediate	WO, client, priority, SLA	No

Email/Teams digests: R2 (the platform replaces the Teams channel rather than feeding it in v1). Quiet hours: none for in-app; business-hours clocks already pause overnight (BR-OBL-001). Delivery failure: the bell inbox is the durable record – pings persist until read.

13. Reporting and insights

Metric / view	Definition	Dimensions	Persona	Refresh	Where it appears
Active WOs	Count of status group = active	Status, client, trade, book	All	Live	KPI row
Awaiting approval	Count in <code>!! waiting for approval</code> + oldest age in days	Client, AM	TL, AM	Live	KPI row
Ready to invoice	Count + total amount in <code>!! ready to invoice</code>	Client, entity	AR, Exec	Live	KPI row
Margin / avg profit	BR-FIN-003...004 over invoiced WOs	Entity, client, trade	Exec, Finance	Live	KPI row
Pulse columns	Obligations by tier (BR-OBL-003) per user	Rule, subject	All	Live	Pulse

Metric / view	Definition	Dimensions	Persona	Refresh	Where it appears
Grey-flag aging	Days a finished WO has sat unreleased	Reviewer, client	HAR	Live	Receivables audit
Collection aging	Days since invoice for unpaid WOs	Client, entity	AR	Live	Receivables invoicing

Finance reconciliation: invoiced/paid figures must match the invoicing records AR keeps today (manual reconciliation at pilot; QuickBooks sync at R2).

14. Non-functional requirements

ID	Category	Requirement	Rationale
NFR-WRK-001	Availability	Available during business hours (Mon-Fri 08:00-18:00 CT) with $\geq 99.5\%$ uptime; emergencies arrive 24x7, so overnight outages must never lose an intake clock	Dispatch and the 2-hour emergency clock depend on it
NFR-WRK-002	Auditability	Every change attributable to a principal, timestamped, immutable (BR-IAM-002)	Client disputes and internal trust tiers
NFR-WRK-003	Correctness of clocks	Obligation clocks accurate to the minute across DST transitions in America/Chicago	The whole obligation model rests on them
NFR-WRK-004	Capacity	100 concurrent internal users; 25,000 work orders live + historical without degradation	Current workspace scale plus growth
NFR-WRK-005	Usability	A new starter can execute SOP-OPS-001 end-to-end without asking a question; gated actions always explain themselves (locked, not hidden)	Training cost; trust-tier onboarding
NFR-WRK-006	Money display	All monetary figures display with tabular numerals and consistent formatting	Misread money is the costliest UX bug
NFR-WRK-007	Visibility safety	Zero paths by which internal content reaches a client surface (BR-ACC-001...003)	One leak is a client-relationship incident

15. Edge cases and exception handling

#	Scenario	Expected behaviour	Rule
E1	Client approves on-site, no CMMS approval will ever come	On-site approval recorded with grantor; wait skipped; no follow-up clock	BR-WRK-009
E2	Emergency WO — quote would slow response	Dispatch may precede quoting; the 2-hour acknowledgment clock still applies; stage order for emergencies is Q5	BR-OBL-007
E3	Tech no-shows the fulfilment visit	Status returns to scheduling; schedule-owed clock restarts; thread records the no-show; vendor note kept	BR-OBL-010
E4	WO cancelled after parts ordered / payment already made	Cancellation recorded with reason; incurred cost remains on the WO's money position; commercial recovery is Q9	BR-WRK-008, BR-FIN-002
E5	Quote rejected after the client already approved on-site	New revision required; on-site approval stands for scope, not price — re-approval needed if total changes (policy to confirm, Q6)	BR-QUO-001
E6	Duplicate payment request for the same visit	Prior requests are surfaced at creation; AP refuses the duplicate at decision	FR-PAY-002, BR-PAY-002

#	Scenario	Expected behaviour	Rule
E7	WO has incurred work but the client declines all options	WO closes as <code>done/incurred</code> ; incurred billing path is Q2	BR-QUO-002
E8	Return trip needed after final soft close	Status <code>return trip needed</code> ; soft-close checks re-open; audit will not accept the WO meanwhile	BR-WRK-005, BR-REC-001
E9	Deliverables store (SharePoint) unreachable during audit	Affected checks report "offline – skipped"; reviewer decides on the remaining evidence; release gate still applies	BR-REC-002
E10	Snooze expires with the obligation still unmet	Obligation returns at its current tier; it never disappears	BR-OBL-004... 005

16. Risks

ID	Risk	Likelihood	Impact	Mitigation	Owner
RSK-WRK-001	Pilot users fall back to ClickUp out of habit and the pilot measures nothing	Medium	High	R1 gate is explicit: one week of intake with zero fallback; Pulse gives daily reasons to stay in	Jordan Brown
RSK-WRK-002	Corrigo/ServiceChannel APIs prove infeasible and intake stays manual	Medium	Medium	Ecotrak first; OA manual intake remains a first-class path, not a workaround	Jordan Brown
RSK-WRK-003	Obligation clocks mistrusted after one wrong fire, and Pulse gets ignored	Low	High	Clock rules published in the catalog; every ping shows its arithmetic; snooze-with-reason keeps humans in charge	Jordan Brown
RSK-WRK-004	Quote gates block work when no ATL is available	Low	Medium	Approval is pooled across ATL/TL/AM/admin; 4-hour clock escalates	Jordan Brown
RSK-WRK-005	Audit rules stricter than practice – release grinds	Medium	Medium	Waiver path for minors (BR-REC-010); rule thresholds reviewed after first month's findings	Head of AR

17. Rollout and change management

Phase	Scope	Duration	Entry criteria	Exit criteria	Rollback
Prototype validation (R0)	Leadership + one OA/OM/AR walk the demo	Done	–	"This is our process" confirmed	–
Pilot (R1)	~5 named users run "Incoming WOs"	~6 weeks of sprints + 1 gate week	Must-requirements pass; SOP-OPS-001 published; personas trained	One week of intake, zero ClickUp fallback	Per-WO reversion to the legacy process (SOP-OPS-001 §8.1); never edit records by hand
Cutover (R2)	Team-by-team waves, all WOs	Phased	Pilot exit met; import reconciled (21K WOs, counts match)	ClickUp Vista space read-only → decommissioned; Teams channel retired	Wave-by-wave hold; ClickUp remains readable (BR-INT-001)

Training: per-role walkthrough of SOP-OPS-001 sections (OA §7 steps 1–3, dispatcher 4–12, ATL 6–7, AP 12–13, AR 14–15); competency = supervised handling of live WOs per SOP §12. **Communications:** clients and techs notice nothing at pilot (their channels are unchanged); AM tells clients before R2 outbound-push changes anything they see. **Data migration:** none at R1 (pilot runs on new intake). R2: one-time ClickUp import with count reconciliation, validated by OA + HAR.

18. Definition of done

- ☐ All **Must** requirements pass their acceptance criteria
- ☐ Every referenced business rule has automated test coverage, including each obligation clock across a weekend and a DST boundary
- ☐ Permissions verified per persona and per record scope, including both sides of every gate (403 + locked control)
- ☐ Zero-path verification that internal content cannot reach a client surface (BR-ACC-001...003)
- ☐ Each integration's failure path demonstrated, not just its success path (§11)
- ☐ SOP-OPS-001 written, reviewed, published
- ☐ Pilot personas trained; competency confirmed
- ☐ M1–M5 instrumentation live and reporting a baseline
- ☐ Known-issues list briefed to whoever supports the pilot
- ☐ Rollback (per-WO reversion) tested, not merely documented

19. Open questions

#	Question	Blocks	Owner	Due	Status
Q1	Accept/decline criteria — trade coverage, geography, NTE floor? Where is a decline recorded today?	FR-WRK-002 refinement	Jordan Brown	Pilot start	Open
Q2	When a client declines all options, is incurred work billed, absorbed, or negotiated?	E7; BR-QUO-002 boundary	Jordan Brown + Finance	R1	Open
Q3	Which specific actions differ between probation / standard / senior within OM and AR?	§9 tier gates; R1 RBAC	Jordan Brown	R1 RBAC build	Open
Q4	May an ATL approve a quote they built themselves? (Conflict register C1)	§10.3	Jordan Brown	R1	Open
Q5	Do emergency WOs formally skip stages (dispatch before quote)?	E2; SOP decision points	Jordan Brown	Pilot start	Open
Q6	Does an on-site approval bind price or only scope — is re-approval needed if the quote total changes?	E5	Jordan Brown	R1	Open
Q7	Retention policy for WOs, money records, and PII (tech names/phones)	§8.3	Jordan Brown + Finance	R2	Open
Q8	Is a recorded payment reversible in-system (clawback), or corrected outside?	§10.3	AP lead	R1	Open
Q9	Commercial recovery when a WO cancels after cost was incurred	E4	Finance	R2	Open
Q10	Tech payment terms: per visit or on completion? Are assessment-only visits paid? Who signs off besides AP?	BR-PAY-004 refinement	AP lead + Jordan Brown	Pilot start	Open
Q11	Definitive payment-method list (replaces free text)	FR-PAY-001	AP lead	Build	Open

#	Question	Blocks	Owner	Due	Status
Q12	Does AP also settle parts/materials invoices (the <code>ap ordered</code> tag suggests yes)?	BR-FIN-002 scope	AP lead	R1	Open
Q13	Per-CMMS differences in the outbound TL Rep push (Ecotrak vs Corrigo vs ServiceChannel)	§11 adapters	Jordan Brown	R2 spike	Open

This PRD may enter review with open questions. It must not enter build with Q1, Q5, Q10, or Q11 unresolved – they block Must requirements.

20. Appendix

20.1 Glossary

Platform-wide terms live in `BUSINESS-RULES-CATALOG-v1.md` §4. PRD-specific: **Book** – a dispatcher’s set of owned WOs (one home book per WO). **Pulse** – the obligations triage view. **Grey flag** – the completion-audit hold before invoicing. **TL Rep** – the curated outbound push to a client’s CMMS.

20.2 Revision history

Version	Date	Author	Change	Re-approval needed
0.1	2026-08-07	J. Brown / Claude (drafted)	Initial draft from product docs + validated prototype behaviour	–